Phase 2: Technology Stack & Architectural Rationale

Selected Technologies & Infrastructure Tools

1. Technology Selection Matrix

Layer	Technology Selected	Version	Purpose & Architectural Justification
Data Processing	Python / Pandas / NumPy	3.14 / 3.1 / 2.5	High-performance tabular data cleaning & feature engineering
Static Visualization	Matplotlib / Seaborn	3.11 / 0.13	Publication-grade figure generation for reports & EDA
Visual BI	Tableau Desktop / Public	v2026.2 (18.1 XML)	Interactive worksheets, calculated fields, dashboards & story
Web Backend	Flask	3.1.3	Lightweight WSGI web server and asynchronous REST API
Web Frontend	Bootstrap 5 / Chart.js	5.3.2 / 4.4.1	Responsive clinical UI, animated canvas graphs & client interactivity
Testing	Pytest	9.2.0	Automated unit test suite validating 11 test assertions
Document Engine	ReportLab	5.1.0	Automated programmatic PDF generation for phase deliverables